

Newman-Penrose Formalism Calculations

Stephani et al., Equation (37.45)

Metric

$$ds^2 = -\frac{9 \left(\frac{\partial V}{\partial t}\right)^2}{\Theta^2 V^2} dt \otimes dt + V dx \otimes dx + V dy \otimes dy + V dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Metric Functions

$$\begin{aligned} V(t, x, y, z) = & \frac{x^2 C(t)^2}{4 V_0(t)} + \frac{y^2 C(t)^2}{4 V_0(t)} + \frac{z^2 C(t)^2}{4 V_0(t)} - \frac{x^2 \Theta(t)^2}{36 V_0(t)} - \frac{y^2 \Theta(t)^2}{36 V_0(t)} - \frac{z^2 \Theta(t)^2}{36 V_0(t)} - \frac{x C(t)^2 x_0(t)}{2 V_0(t)} + \frac{x \Theta(t)^2 x_0(t)}{18 V_0(t)} + \frac{C(t)^2 x_0(t)^2}{4 V_0(t)} - \frac{\Theta(t)^2 x_0(t)^2}{36 V_0(t)} \\ & - \frac{y C(t)^2 y_0(t)}{2 V_0(t)} + \frac{y \Theta(t)^2 y_0(t)}{18 V_0(t)} + \frac{C(t)^2 y_0(t)^2}{4 V_0(t)} - \frac{\Theta(t)^2 y_0(t)^2}{36 V_0(t)} - \frac{z C(t)^2 z_0(t)}{2 V_0(t)} + \frac{z \Theta(t)^2 z_0(t)}{18 V_0(t)} + \frac{C(t)^2 z_0(t)^2}{4 V_0(t)} - \frac{\Theta(t)^2 z_0(t)^2}{36 V_0(t)} + V_0(t) \end{aligned}$$

$$\Theta(t) = \Theta$$

Null Tetrad

Covariant components

$$l = \left(\frac{3 \sqrt{2} \frac{\partial}{\partial t} V}{2 \Theta(t) V} \right) dt + \left(\frac{1}{2} \sqrt{2} \sqrt{V} \right) dx$$

$$n = \left(\frac{3 \sqrt{2} \frac{\partial}{\partial t} V}{2 \Theta(t) V} \right) dt + \left(-\frac{1}{2} \sqrt{2} \sqrt{V} \right) dx$$

$$m = \left(\frac{1}{2} \sqrt{2} \sqrt{V} \right) dy + \left(\frac{1}{2} i \sqrt{2} \sqrt{V} \right) dz$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} \sqrt{V} \right) dy + \left(-\frac{1}{2} i \sqrt{2} \sqrt{V} \right) dz$$

Contravariant components

$$l = \left(-\frac{\sqrt{2} \Theta(t) V}{6 \frac{\partial}{\partial t} V} \right) dt + \left(\frac{\sqrt{2}}{2 \sqrt{V}} \right) dx$$

$$n = \left(-\frac{\sqrt{2} \Theta(t) V}{6 \frac{\partial}{\partial t} V} \right) dt + \left(-\frac{\sqrt{2}}{2 \sqrt{V}} \right) dx$$

$$m = \left(\frac{\sqrt{2}}{2 \sqrt{V}} \right) dy + \left(\frac{i \sqrt{2}}{2 \sqrt{V}} \right) dz$$

$$\bar{m} = \left(\frac{\sqrt{2}}{2 \sqrt{V}} \right) dy + \left(-\frac{i \sqrt{2}}{2 \sqrt{V}} \right) dz$$

Directional Derivatives

$$D = -\frac{\sqrt{2} \Theta(t) V}{6 \frac{\partial}{\partial t} V} \partial_t + \frac{\sqrt{2}}{2 \sqrt{V}} \partial_x$$

$$\Delta = -\frac{\sqrt{2} \Theta(t) V}{6 \frac{\partial}{\partial t} V} \partial_t - \frac{\sqrt{2}}{2 \sqrt{V}} \partial_x$$

$$\delta = \frac{\sqrt{2}}{2 \sqrt{V}} \partial_y + \frac{i \sqrt{2}}{2 \sqrt{V}} \partial_z$$

$$\bar{\delta} = \frac{\sqrt{2}}{2\sqrt{V}} \partial_y - \frac{i\sqrt{2}}{2\sqrt{V}} \partial_z$$

Spin Coefficients

$$\rho = \frac{1}{12} \sqrt{2} \Theta(t) - \frac{\sqrt{2} \frac{\partial}{\partial x} V}{4 V^{\frac{3}{2}}}$$

$$\sigma = 0$$

$$\kappa = -\frac{\sqrt{2} \frac{\partial^2}{\partial t \partial y} V}{4 \sqrt{V} \frac{\partial}{\partial t} V} - \frac{i \sqrt{2} \frac{\partial^2}{\partial t \partial z} V}{4 \sqrt{V} \frac{\partial}{\partial t} V} + \frac{3 \sqrt{2} \frac{\partial}{\partial y} V}{8 V^{\frac{3}{2}}} + \frac{3i \sqrt{2} \frac{\partial}{\partial z} V}{8 V^{\frac{3}{2}}}$$

$$\mu = -\frac{1}{12} \sqrt{2} \Theta(t) - \frac{\sqrt{2} \frac{\partial}{\partial x} V}{4 V^{\frac{3}{2}}}$$

$$\lambda = 0$$

$$\nu = \frac{\sqrt{2} \frac{\partial^2}{\partial t \partial y} V}{4 \sqrt{V} \frac{\partial}{\partial t} V} - \frac{i \sqrt{2} \frac{\partial^2}{\partial t \partial z} V}{4 \sqrt{V} \frac{\partial}{\partial t} V} - \frac{3 \sqrt{2} \frac{\partial}{\partial y} V}{8 V^{\frac{3}{2}}} + \frac{3i \sqrt{2} \frac{\partial}{\partial z} V}{8 V^{\frac{3}{2}}}$$

$$\alpha = -\frac{\sqrt{2} \frac{\partial}{\partial y} V}{8 V^{\frac{3}{2}}} + \frac{i \sqrt{2} \frac{\partial}{\partial z} V}{8 V^{\frac{3}{2}}}$$

$$\beta = \frac{\sqrt{2} \frac{\partial}{\partial y} V}{8 V^{\frac{3}{2}}} + \frac{i \sqrt{2} \frac{\partial}{\partial z} V}{8 V^{\frac{3}{2}}}$$

$$\tau = -\frac{\sqrt{2} \frac{\partial^2}{\partial t \partial y} V}{4 \sqrt{V} \frac{\partial}{\partial t} V} - \frac{i \sqrt{2} \frac{\partial^2}{\partial t \partial z} V}{4 \sqrt{V} \frac{\partial}{\partial t} V} + \frac{\sqrt{2} \frac{\partial}{\partial y} V}{8 V^{\frac{3}{2}}} + \frac{i \sqrt{2} \frac{\partial}{\partial z} V}{8 V^{\frac{3}{2}}}$$

$$\pi = \frac{\sqrt{2} \frac{\partial^2}{\partial t \partial y} V}{4 \sqrt{V} \frac{\partial}{\partial t} V} - \frac{i \sqrt{2} \frac{\partial^2}{\partial t \partial z} V}{4 \sqrt{V} \frac{\partial}{\partial t} V} - \frac{\sqrt{2} \frac{\partial}{\partial y} V}{8 V^{\frac{3}{2}}} + \frac{i \sqrt{2} \frac{\partial}{\partial z} V}{8 V^{\frac{3}{2}}}$$

$$\epsilon = -\frac{1}{24} \sqrt{2} \Theta(t) + \frac{\sqrt{2} \frac{\partial^2}{\partial t \partial x} V}{4 \sqrt{V} \frac{\partial}{\partial t} V} - \frac{\sqrt{2} \frac{\partial}{\partial x} V}{4 V^{\frac{3}{2}}}$$

$$\gamma = \frac{1}{24} \sqrt{2} \Theta(t) + \frac{\sqrt{2} \frac{\partial^2}{\partial t \partial x} V}{4 \sqrt{V} \frac{\partial}{\partial t} V} - \frac{\sqrt{2} \frac{\partial}{\partial x} V}{4 V^{\frac{3}{2}}}$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{\Theta(t)V\frac{\partial}{\partial t}\Theta(t)}{36\frac{\partial}{\partial t}V} + \frac{\frac{\partial^3}{\partial t(\partial y)^2}V}{4V\frac{\partial}{\partial t}V} + \frac{\frac{\partial^3}{\partial t(\partial z)^2}V}{4V\frac{\partial}{\partial t}V} + \frac{\frac{\partial^2}{\partial t\partial x}V\frac{\partial}{\partial x}V}{4V^2\frac{\partial}{\partial t}V} - \frac{\frac{\partial^2}{(\partial x)^2}V}{4V^2} - \frac{\frac{\partial^2}{\partial t\partial y}V\frac{\partial}{\partial y}V}{2V^2\frac{\partial}{\partial t}V} + \frac{9\frac{\partial}{\partial y}V^2}{16V^3} - \frac{3\frac{\partial^2}{(\partial y)^2}V}{8V^2} - \frac{\frac{\partial^2}{\partial t\partial z}V\frac{\partial}{\partial z}V}{2V^2\frac{\partial}{\partial t}V} + \frac{9\frac{\partial}{\partial z}V^2}{16V^3} - \frac{3\frac{\partial^2}{(\partial z)^2}V}{8V^2}$$

$$\Phi_{01} = -\frac{\frac{\partial^3}{\partial t \partial x \partial y} V}{4V \frac{\partial}{\partial t} V} - \frac{i \frac{\partial^3}{\partial t \partial x \partial z} V}{4V \frac{\partial}{\partial t} V} + \frac{3 \frac{\partial^2}{\partial t \partial y} V \frac{\partial}{\partial x} V}{8V^2 \frac{\partial}{\partial t} V} + \frac{3i \frac{\partial^2}{\partial t \partial z} V \frac{\partial}{\partial x} V}{8V^2 \frac{\partial}{\partial t} V} + \frac{\frac{\partial^2}{\partial x \partial y} V}{8V^2} + \frac{i \frac{\partial^2}{\partial x \partial z} V}{8V^2} + \frac{3 \frac{\partial^2}{\partial t \partial x} V \frac{\partial}{\partial y} V}{8V^2 \frac{\partial}{\partial t} V} - \frac{9 \frac{\partial}{\partial x} V \frac{\partial}{\partial y} V}{16V^3} + \frac{3i \frac{\partial^2}{\partial t \partial x} V \frac{\partial}{\partial z} V}{8V^2 \frac{\partial}{\partial t} V} - \frac{9i \frac{\partial}{\partial x} V \frac{\partial}{\partial z} V}{16V^3}$$

$$\Phi_{02} = -\frac{\frac{\partial^3}{\partial t(\partial y)^2} V}{4V\frac{\partial}{\partial t} V} - \frac{i\frac{\partial^3}{\partial t\partial y\partial z} V}{2V\frac{\partial}{\partial t} V} + \frac{\frac{\partial^3}{\partial t(\partial z)^2} V}{4V\frac{\partial}{\partial t} V} + \frac{3\frac{\partial^2}{\partial t\partial y} V\frac{\partial}{\partial y} V}{4V^2\frac{\partial}{\partial t} V} + \frac{3i\frac{\partial^2}{\partial t\partial z} V\frac{\partial}{\partial y} V}{4V^2\frac{\partial}{\partial t} V} - \frac{9\frac{\partial}{\partial y} V^2}{16V^3} + \frac{\frac{\partial^2}{(\partial y)^2} V}{8V^2} + \frac{i\frac{\partial^2}{\partial y\partial z} V}{4V^2} + \frac{3i\frac{\partial^2}{\partial t\partial y} V\frac{\partial}{\partial z} V}{4V^2\frac{\partial}{\partial t} V} - \frac{3\frac{\partial^2}{\partial t\partial z} V\frac{\partial}{\partial z} V}{4V^2\frac{\partial}{\partial t} V} - \frac{9i\frac{\partial}{\partial y} V\frac{\partial}{\partial z} V}{8V^3} + \frac{9\frac{\partial}{\partial z} V^2}{16V^3} - \frac{\frac{\partial^2}{(\partial z)^2} V}{8V^2}$$

$$\Phi_{10} = -\frac{\frac{\partial^3}{\partial t \partial x \partial y} V}{4V \frac{\partial}{\partial t} V} + \frac{i \frac{\partial^3}{\partial t \partial x \partial z} V}{4V \frac{\partial}{\partial t} V} + \frac{3 \frac{\partial^2}{\partial t \partial y} V \frac{\partial}{\partial x} V}{8V^2 \frac{\partial}{\partial t} V} - \frac{3i \frac{\partial^2}{\partial t \partial z} V \frac{\partial}{\partial x} V}{8V^2 \frac{\partial}{\partial t} V} + \frac{\frac{\partial^2}{\partial x \partial y} V}{8V^2} - \frac{i \frac{\partial^2}{\partial x \partial z} V}{8V^2} + \frac{3 \frac{\partial^2}{\partial t \partial x} V \frac{\partial}{\partial y} V}{8V^2 \frac{\partial}{\partial t} V} - \frac{9 \frac{\partial}{\partial x} V \frac{\partial}{\partial y} V}{16V^3} - \frac{3i \frac{\partial^2}{\partial t \partial x} V \frac{\partial}{\partial z} V}{8V^2 \frac{\partial}{\partial t} V} + \frac{9i \frac{\partial}{\partial x} V \frac{\partial}{\partial z} V}{16V^3}$$

$$\Phi_{11} = -\frac{\Theta(t)V\frac{\partial}{\partial t}\Theta(t)}{72\frac{\partial}{\partial t}V} + \frac{\frac{\partial^3}{\partial t(\partial x)^2}V}{4V\frac{\partial}{\partial t}V} - \frac{5\frac{\partial^2}{\partial t\partial x}V\frac{\partial}{\partial x}V}{8V^2\frac{\partial}{\partial t}V} + \frac{9\frac{\partial}{\partial x}V^2}{16V^3} - \frac{\frac{\partial^2}{(\partial x)^2}V}{4V^2} + \frac{\frac{\partial^2}{\partial t\partial y}V\frac{\partial}{\partial y}V}{8V^2\frac{\partial}{\partial t}V} - \frac{\frac{\partial^2}{(\partial y)^2}V}{8V^2} + \frac{\frac{\partial^2}{\partial t\partial z}V\frac{\partial}{\partial z}V}{8V^2\frac{\partial}{\partial t}V} - \frac{\frac{\partial^2}{(\partial z)^2}V}{8V^2}$$

$$\Phi_{12} = \frac{\frac{\partial^3}{\partial t \partial x \partial y} V}{4 V \frac{\partial}{\partial t} V} + \frac{i \frac{\partial^3}{\partial t \partial x \partial z} V}{4 V \frac{\partial}{\partial t} V} - \frac{3 \frac{\partial^2}{\partial t \partial y} V \frac{\partial}{\partial x} V}{8 V^2 \frac{\partial}{\partial t} V} - \frac{3 i \frac{\partial^2}{\partial t \partial z} V \frac{\partial}{\partial x} V}{8 V^2 \frac{\partial}{\partial t} V} - \frac{\frac{\partial^2}{\partial x \partial y} V}{8 V^2} - \frac{i \frac{\partial^2}{\partial x \partial z} V}{8 V^2} - \frac{3 \frac{\partial^2}{\partial t \partial x} V \frac{\partial}{\partial y} V}{8 V^2 \frac{\partial}{\partial t} V} + \frac{9 \frac{\partial}{\partial x} V \frac{\partial}{\partial y} V}{16 V^3} - \frac{3 i \frac{\partial^2}{\partial t \partial x} V \frac{\partial}{\partial z} V}{8 V^2 \frac{\partial}{\partial t} V} + \frac{9 i \frac{\partial}{\partial x} V \frac{\partial}{\partial z} V}{16 V^3}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type " I "